

Lab 05 - Autonomous Vehicles

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Abstract

This report outlines the automotive industry's evolution from early mass production to the modern era of electric and autonomous mobility. It reviews key milestones in safety and environmental impact, such as the introduction of seatbelts and electric vehicles.

Additionally, the text defines autonomous driving levels 3 through 5 and addresses technical challenges regarding infrastructure and stochastic driving processes. Finally, it explores the philosophical drivers behind automation versus the human desire to drive.

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1 Introduction

This report summarizes the major milestones in the automotive industry, tracing its progression from early commercialization to the current frontier of autonomous driving, and concluding with an examination of the driving force behind this technological advancement. Professor Mehdi Ahmadian from Virginia Institute of Technology guided us through these steps.

2 Milestones in automotive history

Commercial production began with the Ford Model T, which was a revolutionary vehicle at the time. Its production on the assembly line marked a significant social and economic development, making the automobile available and affordable, facilitating the growth of cities as people could move farther from factories.

Early foundational technical achievements included Carl Benz first automobile, Porsche first hybrid car, and Rolls Royce first luxury vehicle. Design and functionality were expanded by the Citroen first Front-Wheel Drive (FWD). The Jeep was noted as the first offroad vehicle, while the Ford Mustang served as the first sport car targeted at young consumers.

In the second part of the century, some important steps were made also in terms of safety and the environmental impact of vehicles started to be taken into account. The introduction of the Volvo three-point seatbelt and the public safety concerns exposed in book *"Unsafe at Any Speed"* were the most significant moments. Subsequently, General Motors introduced the first airbag. The *"Clean Air Act"* drew attention to emissions, which led to General Motors introducing the first electric vehicle, though its adoption was hindered by poor charging infrastructure. This paved the way for the Toyota first hybrid electric vehicle, which featured high-voltage batteries designed to last the full life of the vehicle. The era of modern electric mobility was solidified by the Tesla first fully electric mass-manufactured vehicle.

Tesla was also the first manufacturer to strongly believe in autonomous driving; this lead to the second part of the discussion: the state of art of self-driving cars.

3 State of art of autonomous driving

Vehicle autonomy is classified into various levels:

- **Level 3:** The car performs certain tasks autonomously, but the human driver must remain present and ready to intervene.
- **Level 4:** The car is capable of full control of the driving process, though the driver is still present and able to take over if necessary.
- **Level 5:** Full automation is achieved. The driver is no longer required to be present, and the steering wheel may be eliminated. Level 5 vehicles are intended to think, behave, and interact like human drivers, with the predicted benefit of increased productivity or leisure time for occupants.

The current technological challenge is defined by two major impediments: the lack of necessary infrastructure for vehicles to interact with their environment, and the fact that driving is an entirely stochastic (random and unpredictable) process, which severely complicates reliable software implementation.

4 Philosophical aspect of self-driving cars

Is the development of autonomous driving being driven by the availability of technology (Technological Capability → Autonomous Vehicles), or is the core desire for autonomous mobility (safety, productivity) demanding that the requisite technologies be implemented (Autonomous Vehicles → Technological Implementation)?

Furthermore, the adoption of Level 5 autonomy must contend with a significant factor: some individuals enjoy driving. Both vehicle safety and the pursuit of climate change solutions were cited as parameters that strongly influence the direction of technological development, suggesting that a balance between needs and pleasure must be found.